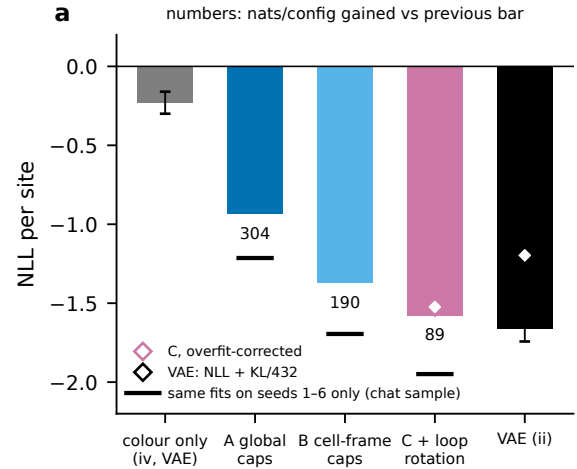
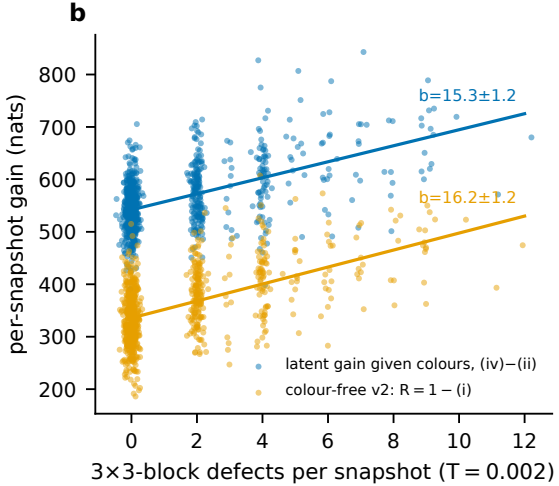
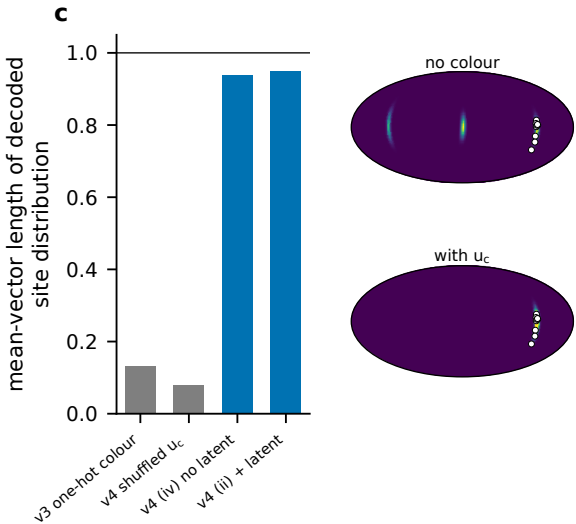






d

channel	information G	captured by VAE?
colouring (3-colour pattern)	≤ ln 3 per site (conditioning input)	only via gauge-free input u_c (v4)
global plane + triad (A)	304 nats/cfg	yes ($R = 1$, $l = 3$ latent)
local frame (B)	190 nats/cfg (~12 per 3x3 cell)	partly; not isolated
loop rotation (C)	~1.4 nats/loop	not resolved ($a = 0.16 \pm 0.04$)
phase defects	~15–23 nats/defect	yes ($b > 0$ in all models)
gauge bits (time reversal, triad branch)	$G \le \ln 2$ (assumed)	not measured here (assumed not encoded)